

SOHAM TILEKAR

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PROFILE

Software Engineering student at UNSW working across experimental research and engineering — from testing a physics theory for a science fair to building sensor-driven robots

I design experiments, analyze data, and build what's needed to support them: CAD, electronics, and code

EDUCATION

Bachelor of Software Engineering (Honours)
University of New South Wales

September 2025 - September 2028

- WAM: 75.33 (Distinction)

RELEVANT COURSEWORK

- Programming (COMP1511, COMP1521, COMP2521, COMP1531, COMP3311)
- Engineering (DESN1000, DESN2000)
- Mathematics (MATH1081, MATH2400)
- Credits Transferred for MATH1231 and MATH1131 (In Progress)

RESEARCH EXPERIENCE

Student Researcher

April 2020 - October 2020

Experimental Verification of the Mpemba Effect

- Explored Practical Use of the Mpemba Effect in Pre-cooling Harvested Vegetables
- Designed and ran controlled experiments to verify the Mpemba effect, contributing equally across hypothesis design, data collection, and analysis with three teammates
- Applied experimental findings to propose a practical pre-cooling method for harvested vegetables, translating a physics phenomenon into an agricultural application
- Presented experimental methodology and findings at a science fair, communicating results to a public and academic audience

PROJECTS & AWARDS

ENVIROBOT - 1st Place (\$250 prize)

2026

CREATE UNSW & Student Energy - robotics and case competition (on-field testing + case study), 4-member team

- Designed and manufactured robot chassis and sensor mount using AutoCAD, engineering a compact tank-style platform with a servo-actuated arm to lower a soil sensor to ground level
- Assembled and soldered the PCB and coded a sensor system, achieving 11/12 correct data readings during on-field testing

Closet Ninja - 3rd Place (\$200 prize)

2026

Computer Science and Engineering society's Flagship Hackathon

- Trained and integrated an OpenAI vision model and a YOLO model to detect and classify clothing items from a fashion dataset
- Converted identified clothing items into image embeddings and storing to enable later selfies to be matched against a user's existing wardrobe
- Built pipeline to track clothing usage over time, generating insights into a user's wardrobe habits

EXPERIENCE

Part-Time Chess Instructor

December 2025 - Present

Victorious Chess Academy

- Coach a group of 20-25 students per three-month term, creating lesson plans improving students' playing strength over each cycle
- Maintain a personal rating of 2300 ELO on Chess.com, drawing on competitive-level experience to inform instruction

VOLUNTEER EXPERIENCE

General Volunteer

June 2026 - Present

Arc - UNSW Bike-ology

- Repaired and maintained donated bikes, restoring bikes to safe, rideable condition for the campus community
- Ran outreach efforts, promoting the initiative and encouraging student engagement with sustainable transport on campus

CERTIFICATIONS

CS50: Introduction to Computer Science - Harvard University

- Harvard's intro course covering algorithms, data structures, and problem-solving in C, Python

CS50 SQL: Introduction to Databases with SQL - Harvard University

- A focused follow-up course on relational database design, querying, and optimization using SQL

SKILLS

- Programming Languages: C, C++, Python, JavaScript
- Tools & Platforms: Git/GitHub, Linux/Unix-based systems, AutoCAD
- Databases: SQL